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Application Paper

Evaluating machine learning-based limited-area weather forecasting models for wind energy applications

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Abstract

Machine learning-based limited-area models (LAMs) have recently shown that they are capable of outperforming their conventional, numerical counterparts for generating high-resolution weather forecasts. The suitability of these models for renewable energy applications, for which accurate, high-resolution weather forecasts are vital, remains however largely unexplored. In particular, the implications of the choice of model architecture for forecasting weather variables critical to wind power are not fully understood. In this study, we investigate how architectural design choices in machine learning-based LAMs influence forecast quality at high spatial resolution, with a focus on near-surface and hub-height wind speeds. Using the *Anemoi* framework, we train multiple encoder-processor-decoder architectures within a consistent limited-area setup over a North-West European domain, where we evaluate these models for lead times up to three days. In addition, we provide a small proof-of-concept case study in which these wind forecasts are converted to wind power estimates for the Belgian offshore zone using a data-driven transformer model. The results show that architectural choices lead to notable differences in forecast accuracy, both across lead times and spatially, with more pronounced differences between models at shorter lead times that are less significant for extreme wind speed events. Furthermore, none of the architectures tested emerge as overall best performing method while the model realizing the highest wind speed forecast skill does not necessarily yield superior wind power forecasts.

Impact Statement

Accurate, high-resolution renewable energy forecasts are essential in order to realize the energy transition. Building on the recent advances in high-resolution limited area modeling, this work evaluates how architectural choices in machine learning-based weather prediction models affect forecast quality for variables directly relevant to wind power, and demonstrates how these differences propagate to downstream energy applications. The results show that no single architecture emerges as universally superior, and that the model resulting in the most accurate wind speed forecasts does not necessarily yield the most accurate power forecasts, highlighting the importance of end-to-end evaluation in energy applications.

1. Introduction

Accurate weather forecasts are of vital importance for a wide variety of different sectors, such as the agricultural, transport and public health sector (Challinor and Reading 2004; Mohanty et al 2015; Nurmi et al 2013). In particular, given the weather dependence of solar and wind power production,

they are necessary to realize accurate renewable energy forecasts and as such are crucial for balancing the grid, i.e. matching supply and demand (Van Poecke, Tabari et al 2024). Forecasting wind power is, through its strong dependency on wind speed, a particularly complex problem, given its sensitivity to a wide range of complex flow scales and atmospheric variability (De Freitas et al 2018; Porté-Agel et al 2020). Furthermore, inaccurate predictions of wind speed, and as such wind power, lead to significant economic losses and additional CO2 emissions (Bruninx and Delarue 2014). As power systems increasingly rely on weather-dependent generation, the demand for reliable, high-resolution weather forecasts tailored to renewable energy applications continues to grow (Van Poecke, Tabari et al 2024).

In recent years, the advent of large machine learning-based weather prediction (MLWP) models has fundamentally altered the weather forecasting landscape (Ben Bouallègue et al 2024). However, most existing MLWP models focus on global forecasting at moderate spatial resolutions, largely reflecting the resolution of commonly used training datasets such as ERA5 (Wijnands et al 2025). To overcome this limitation, recent work has looked into extending these MLWP models toward high-resolution regional forecasting which are trained on high-resolution datasets such as the Copernicus European Regional Reanalysis (CERRA) dataset (Ridal et al 2024). In order to do so, two approaches have emerged: limited-area models (LAMs), which produce forecasts on a high-resolution inner region while employing a lower-resolution boundary forcing (Adamov et al 2025; Oskarsson et al 2023; Wijnands et al 2025) and stretched-grid models (SGMs), which provides forecasts at both the global level, at lower resolution, and at the regional level, at higher resolution (Nipen et al 2025; Wijnands et al 2025). Similar to their global counterparts, these regional MLWP models have shown they can outperform conventional numerical weather prediction in all-round weather skill at a lower computational cost (Adamov et al 2025). At the same time, variable-specific research and studies on the influence of internal modeling choices in MLWP are still at an early stage. Initial studies have begun to explore how training objectives, such as the choice of architecture or loss function in graph-based MLWP models, can affect forecast accuracy, bias, and variability across variables, regions, and lead times (Van Poecke, Cisneros et al 2025; Van Poecke, Meuris et al 2024). These results suggest that architectural and training choices may be tailored for specific downstream applications, like renewable energy forecasting.

Building on these developments, this study aims to assess how architectural choices in MLWP models affect forecast quality in a high-resolution limited-area setting, with a specific focus on variables relevant to wind power forecasting. Given the success of the encoder-processor-decoder architecture in graph-based weather forecasting models (Keisler 2022), three different graph neural networks (GNNs), i.e. a standard GNN, GraphTransformer and Transformer, are trained utilizing the *Anemoi* framework. These models are trained on a part of the CERRA domain covering large parts of the United Kingdom, the North Sea and parts of the coastal areas of France, Belgium, the Netherlands and Germany, a region characterized by dense wind energy infrastructure (International Energy Agency 2019). Following Wijnands et al 2025, a large, low resolution boundary is provided as forcing in the form of ECMWF Reanalysis v5 (ERA5) data (Hersbach et al 2020). Given the rather limited size of the region of interest, we opted for the LAM approach, but it should be noted that this research could be extended with a SGM approach, which is a pathway we consider for future work. Performance is evaluated on wind speed at ten and one hundred meter (ws10 and ws100), variables highly relevant to wind power forecast skill. In addition, we include a small proof-of-concept case study in which the resulting wind forecasts are translated into wind power estimates for the Belgian offshore zone, illustrating how differences in weather forecast accuracy may propagate to downstream energy applications. By providing an application-driven comparison of MLWP architectures for high-resolution regional forecasting, this work aims to contribute to model selection for subsequent wind power forecasting systems and renewable energy integration at large.

2. Methodology

2.1. Anemoi

The models and graphs discussed in this work are trained and configured using the *Anemoi* framework developed by ECMWF and various national weather services. *Anemoi* is an open-source Python framework for creating machine learning-based weather forecasting systems. In Anemoi, the dataset is defined on a grid whose points form the nodes of a graph, along which information is propagated by a deep learning model, typically a GNN, to produce forecasts. The main components needed to train a model in *Anemoi* as such are datasets, graphs and models, all three of which are briefly discussed below[1].

2.2. Data

This work employs two datasets: CERRA (Ridal et al 2024) and ERA5[2](Hersbach et al 2020), from which a wide number of surface, pressure, height level and forcing variables are selected. Both datasets entail identical variables, including wind speed and direction, temperature, geopotential height and specific humidity, among others. Details regarding the exact variables, pressure levels and forcings are presented in Table 1 in section B of the Appendix. CERRA operates on a lambert conformal conical grid and has a spatial resolution of 5.5×5.5 km with a temporal resolution of three hours, whereas ERA5 has a resolution of 31×31km on a reduced Gaussian grid, where we utilize the same three hourly resolution as CERRA. Forecasts are made in the selected subregion of CERRA, visualized by the green rectangle on the left of Figure 1, representing the high resolution CERRA data, surrounded by the coarser ERA5 data. The models are trained to forecast meteorological variables in the region of interest using as input the atmospheric state at t and $t - 3h$.

2.3. Graphs

All models considered in this study are implemented as a type of GNN and therefore require an explicit graph representation. The graph constructed here, using the *Anemoi-graphs* package, closely follows the set-up of Wijnands et al 2025. It consists of data nodes, representing physical weather variables on the input grids, and hidden nodes, which form a latent representation on which the model operates. The encoder-graph, shown in Figure 1, connects data nodes to hidden nodes with edges and as such projects the meteorological state onto a spatially lower-dimensional latent space. The right panel of Figure 1 reveals this flow of information in more detail. In the processor, edges connect hidden nodes to each other on different interaction scales, allowing information exchange over various distances. Finally, the decoder maps the processed latent representation back to the data nodes in the regional domain to produce forecasts at the original grid resolution, while excluding the boundary region from predictions. Graph images for the processor and decoder are similar to Figure 1 and are therefore omitted here[3].

2.4. Models

All models trained in this work follow the encoder–processor–decoder paradigm and learn message passing over a graph. As such, they constitute graph neural networks (GNNs) by definition. We trained three different architectures: a standard Graph Neural Network (GNN), a GraphTransformer and a Transformer. The GNN implements non-attentional message passing with incorporation of graph-aware knowledge, i.e. information regarding both edges and nodes. The Transformer, on the other hand, performs self-attention over a local neighborhood defined by a fixed-size sliding window, effectively

[1]More information regarding the *Anemoi* framework can be found in the documentation: https://anemoi.readthedocs.io/en/latest/ or the Github repository https://github.com/ecmwf/anemoi.

[2]Both of which can be freely downloaded from the copernicus website https://cds.climate.copernicus.eu/datasets/.

[3]The implementation of the graph can be found in Github https://github.com/UAntwerpM4S/WindAI/tree/main/graphs

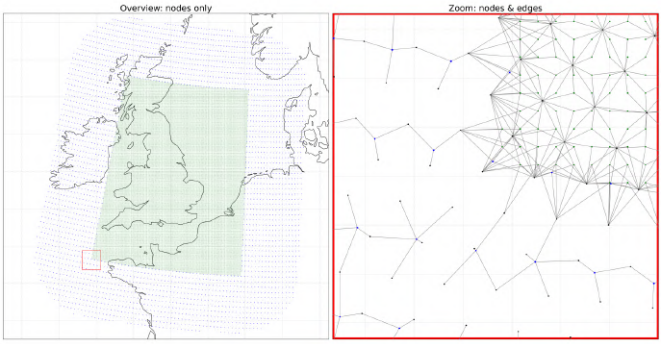


Figure 1: Encoder, where blue and green nodes constitute ERA5 and CERRA data points respectively, whereas hidden nodes are represented by black dots. Data and hidden nodes are connected by encoder edges.

treating the nodes as a sequence rather than an irregular graph. Lastly, the GraphTransformer combines both approaches by performing attentional message passing while being locally graph-aware, i.e. the algorithm employs the attention mechanism to learn the optimal weighing of the information of its neighbors. All three models share the same encoder-processor-decoder backbone with 8 layers in the processor and a feature dimension of 512 channels activated with a Gaussian Error Linear Unit (GELU). They differ mainly in their processor architecture: the GNN employs non-attentional message passing, while the GraphTransformer and Transformer share GraphTransformer-based encoder and decoder mappers and both use 8 attention heads, with the Transformer additionally operating over a fixed sliding window of size 128[4].

2.5. Training

Each model was trained on a single NVIDIA L40S GPU using a two-stage strategy. The initial phase consisted of 150000 steps of single-step training, utilizing the AdamW optimizer with a learning rate of 3.125×10^{-5} and then another 6000 fine tuning steps with learning rate 8.0×10^{-7} (S Lang et al 2024), where the autoregressive depth was increased to 6 steps to minimize error accumulation during recursive inference. During this stage, the mean squared error (MSE) is employed as loss function, incorporating the variable weighting scheme from AIFS (S Lang et al 2024), with the exception of wind speed at ten and one hundred meter, which were given a slightly higher weight of 2 as compared to 1 in AIFS. Training was performed on the period starting at 01/01/2015 and ending at 31/01/2024, while validation was done over the period from 01/02/2024 to 31/07/2024 and lastly the period spanning from 01/08/2024 to 31/07/2025 was used for testing. Training took around 4.7 hours for the Transformer, 16.5 for the GraphTransformer and 25.1 hours for the GNN. The Transformer's significantly shorter training time can be attributed to its use of attention across a fully flattened sliding window, which enables greater parallelization compared to the graph-aware message passing of the GNN and GraphTransformer.

2.6. Evaluation

2.6.1. Wind speed

In order to evaluate the suitability of these models for wind power forecasting, we evaluate wind speed at ten and one hundred meter, corresponding respectively to near-surface conditions and typical offshore hub heights. Forecasts from each model are compared against the CERRA reanalysis, which is treated

[4]The exact implementation can be found on github: https://github.com/UAntwerpM4S/WindAI/tree/main/training/CI.

as reference truth. We calculate the root mean squared error (RMSE) for all three methods for the test period and take the average over the inner region of Figure 1 in order to obtain per lead time values. In order to infer the spatial distribution of the error, we compute RMSE per grid cell averaged over all lead times and initialisation times. Additionally, the capability of the three models to forecast extreme wind speed events is assessed by evaluating forecasts only at timesteps where the CERRA truth exceeds the 95th percentile, which is computed per grid cell over the full verification period. Errors are then averaged spatially to obtain RMSE as a function of lead time. Lastly, we assess the spatial bias by computing the mean difference between forecast and reanalysis across all lead times and initialisation times. These bias figures can be found in Appendix A, together with verification of key meteorological variables, other than wind speed.

2.6.2. Wind Power

To assess how differences in wind speed forecast quality may translate to downstream energy applications, we also provide a short proof of concept regarding wind power forecasting in the Belgian offshore zone (BOZ). Power estimates are derived from the weather forecasts using a transformer built around self-attention. This data-driven model, developed in previous work for the postprocessing of wind speed (Van Poecke, Finn et al 2025), is adapted to apply multi-head self-attention across the lead time dimension to capture temporal dependencies. The transformer is trained using the wind speed and direction forecasts of ten and one hundred meter of the three models described above, for which the first 8 months of the testing data are used, i.e. the period 01/08/2024 - 31/03/2025, whereas the next two months were used for validation (the period 01/04/2025-31/05/2025) and the last two as testing data (the period 01/06/2025-31/07/2025). As ground truth, we employ the publicly available wind power production data from the European Network of Transmission System Operators for Electricity (ENTSO-E)[5]. ENTSO-E is a European association representing 40 Transmission System Operators (TSOs) of 36 European countries with the coordinated operation of the European electricity system as central mission (ENTSO-E 2026). The power forecasts made by the transformer are evaluated by comparing against the values obtained from ENTSO-E, using RMSE across lead times and summed across all the wind farms in the BOZ, as depicted in Figure 5a. For clarity, the transformer utilized for power forecasting is referred to as "PowerTransformer" and the Transformer-based weather model will be specified with "Transformer" in the remainder of this work[6].

3. Results

It should be noted that the overall normalized and weighted MSE values after training and validation were the lowest for the GraphTransformer with 0.17 and 0.22 respectively, which indicates that this architecture realized the most accurate weather forecasts during training and validation. The Transformer, on the other hand, achieved weighted MSE values of 0.19 and 0.23 while requiring less than a third in training time. Lastly, the GNN required significantly more training time to realize higher MSE values, i.e. 0.24 and 0.31.

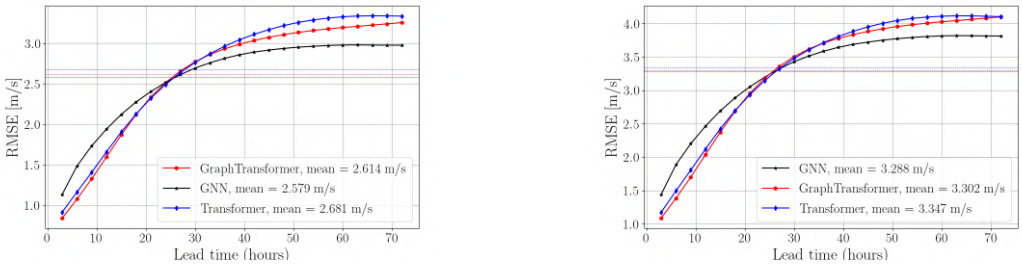
3.1. Wind speed

Figure 2 shows the RMSE per lead time of the wind speed forecasts as generated by the GNN (black), the GraphTransformer (red) and the Transformer (blue). The results for wind speed at both ten (Figure 2a) and one hundred meter (Figure 2b) show very similar trends: overall, the GraphTransformer and Transformer achieve a comparable accuracy per lead time, with the GNN deviating significantly compared to the other two models. The GNN performs worse in early lead times but compensates with more accurate forecasts

[5]The data can be freely downloaded on the ENTSO-E website: https://transparency.entsoe.eu/

[6]The implementation of the PowerTransformer can be found on Github: https://github.com/UAntwerpM4S/WindPowerTransformer

at longer lead times, for both ws10 and ws100. Next, Figure 3 shows the spatial distribution of the RMSE for the three models. For clarity, Figure 3 shows differences in RMSE between models, rather than just absolute values per model, to highlight differences in accuracy per region. Red indicates that the first model (GNN or GraphTransformer) performs worse, blue indicates that the latter (GraphTransformer or Transformer) realizes more accurate forecasts. Again, these figures reveal a similar pattern for ws10 and ws100. Figures 3a, 3b, 3d and 3e show that the GNNs performs worst in the Belgium offshore zone and wider North Sea region, particularly when compared to the GraphTransformer. Differences between the GNN and Transformer appear less significant (Figures 3b and 3e), whereas the GraphTransformer outperforms the Transformer in the Southern part of the region of interest, while the opposite holds in the Northern region (Figures 3c and 3f). Notably, some of these figures, especially Figures 3c and 3f, exhibit some spatial texture that stems from the underlying graph, suggesting that the errors retain some structure inherited from the graph connectivity rather than reflecting purely physical spatial patterns.



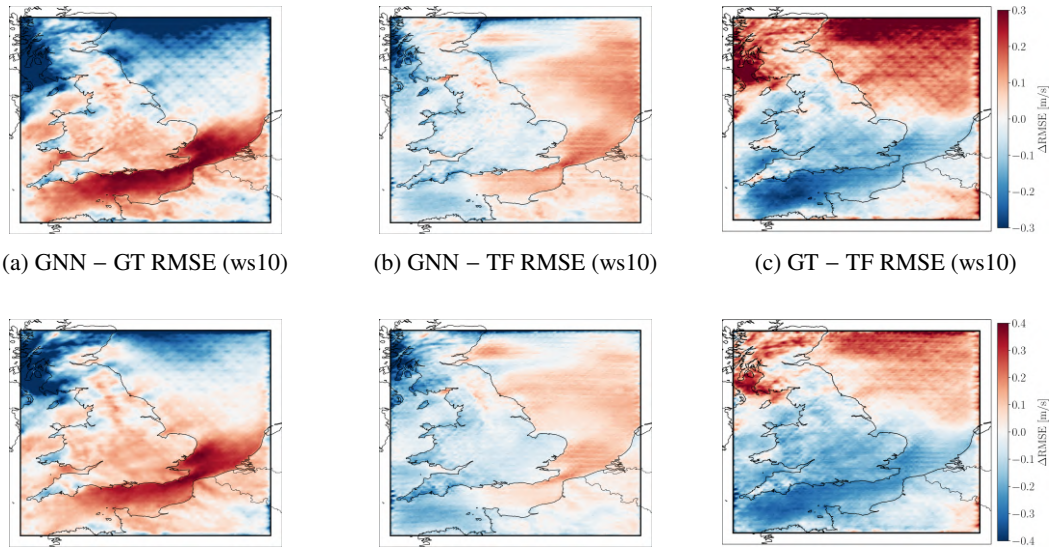
(a) ws10 (b) ws100

Figure 2: RMSE in function of lead time for all three models for wind speed at ten meter (*a*) and one hundred meter (*b*).

Next, Figures 4a and 4b show the performance of the three models on extreme wind speed events, defined as CERRA values exceeding the 95th percentile of the verification period distribution at each grid cell. Forecast errors are accumulated only at those grid cells and timesteps where the CERRA truth exceeds the local threshold, and then averaged spatially to obtain RMSE as a function of lead time. For context, the average threshold per grid cell was 11.018 m/s for ws10 and 14.562 m/s for ws100. The difference in performance between GNN on the one hand and the GraphTransformer and Transformer on the other hand are less articulate here than was the case for the regular RMSE values (Figures 2a and 2b). Nonetheless, the GNN performs worst in the early lead times, with the Transformer model performing best overall.

3.2. Wind Power case study

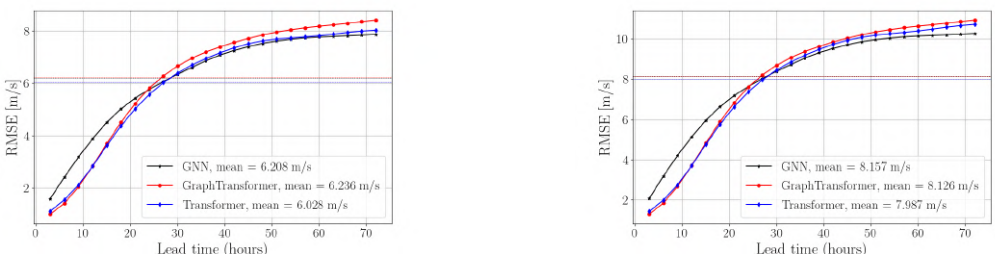
RMSE values of the wind power forecasts are shown in Figure 5b, where each curve represents power forecasts made by the PowerTransformer based on forecasts of the GNN (black), GraphTransformer (red) and Transformer (blue). RMSE is shown as a percentage of the total installed capacity in the BOZ. Interestingly, the results do not follow the same trend as those of the wind speed forecasts as shown in Figures 2a and 2b. The most accurate power forecasts are those generated by the PowerTransformer trained on the GraphTransformer-based wind forecasts, with those originating from the Transformer lagging slightly behind. The GNN-based power forecasts are significantly worse in early lead times,



(a) GNN − GT RMSE (ws10) (b) GNN − TF RMSE (ws10) (c) GT − TF RMSE (ws10)

(d) GNN − GT RMSE (ws100) (e) GNN − TF RMSE (ws100) (f) GT − TF RMSE (ws100)

Figure 3: Spatial distribution of the difference in RMSE for wind speed at 10 m (top row) and 100 m (bottom row), averaged over all lead times. The left and middle column show the difference between the GNN and the GraphTransformer and Transformer respectively; (red: GNN worse, blue: competing model worse), whereas the last column shows the difference between the GraphTransformer and Transformer (red: GraphTransformer worse, blue: Transformer worse).



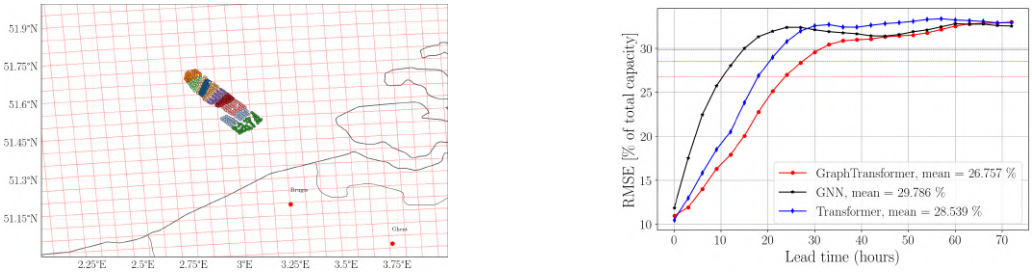
(a) ws10, 95th percentile (b) ws100, 95th percentile

Figure 4: RMSE in function of lead time for all three models for wind speed at ten meter (*a*) and one hundred meter (*b*), verified only for wind speeds higher than the 95th percentile.

which was to be expected from the wind speed analysis above, but do not outperform the GraphTransformer at late lead times in this case. For every lead time, the GraphTransformer-based power forecasts are the most accurate.

4. Discussion and future prospects

This work provides a first, application-oriented comparison of different encoder–processor–decoder architectures for machine learning–based limited-area weather forecasting, with a focus on variables



(a) Wind farms

(b) Wind power

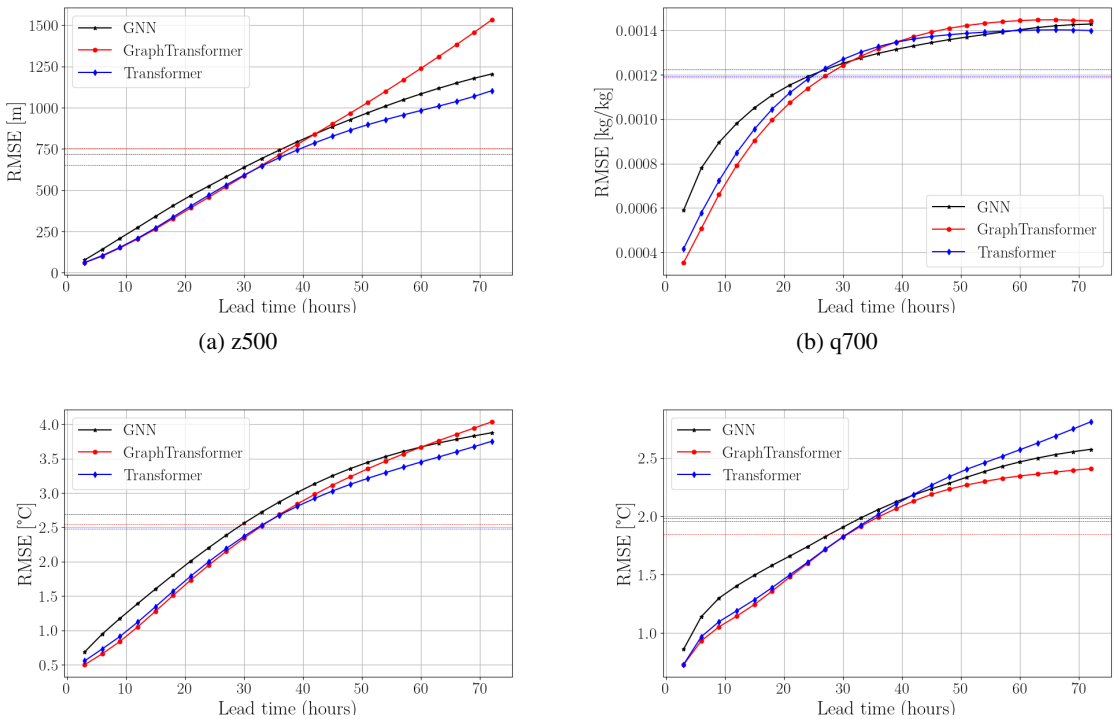
Figure 5: (*a*) Wind farms in the Belgian offshore zone (*b*) and RMSE as a function of lead time for wind power forecasts made by the PowerTransformer based on the wind speed forecasts of the three models.

relevant to wind energy applications. For wind speed at ten and one hundred meter, the GraphTransformer and Transformer achieve comparable accuracy across lead times, while the GNN deviates more substantially: it performs worse at early lead times but compensates with improved forecasts at longer lead times. Spatially, the GNN performs worst in the Belgian offshore zone and the wider North Sea region, particularly relative to the GraphTransformer, while the GraphTransformer and Transformer show a south-north divided performance pattern. For extreme wind speed events, performance differences between models are less pronounced, though the GNN still underperforms at early lead times, with the Transformer performing best overall. The superior performance of the GraphTransformer and Transformer at shorter lead times can likely be attributed to the attention mechanism, which enables both models to dynamically weight the most relevant spatial relationships. At longer lead times this advantage appears to diminish, possibly because the small domain size means weather systems have largely traversed the region within the three-day forecast window, reducing the predictive value of the initial atmospheric state. Interestingly, the ranking of architectures for power forecasting does not mirror that observed for wind speed: the GraphTransformer-based power forecasts achieve the lowest error across all lead times, while those stemming from the Transformer lag slightly behind, and the GNN-based forecasts perform significantly worse at early lead times without recovering to outperform the GraphTransformer at longer lead times. This may be partly explained by the spatial distribution of forecast errors: as shown in Figures 3a, 3b, 3d and 3e, the GNN performs particularly poorly in the Belgian offshore zone, which is precisely the region evaluated in the power case study. These results underscore that the model achieving the highest overall wind speed forecast skill does not necessarily yield superior wind power forecasts.

Several directions for future work follow naturally from this preliminary study. First, training both the weather models and the PowerTransformer on a substantially larger dataset would allow them to achieve higher accuracy and improve robustness while also allowing for a more rigorous investigation of whether the observed lead-time behaviour of the GNN is driven by domain size or by intrinsic architectural limitations. Second, an important next step is to incorporate wind power information in the training process of the weather models, rather than simply using its output to generate power predictions with an external model. Including wind power during training allows the model to tailor its training strategy towards power forecasting and consequently could result in more accurate power predictions. Finally, future studies will include a wider range of wind speed and power observations, both offshore and onshore, as using a more extensive dataset would allow for a more representative assessment of the different forecasting architectures utilized.

A. Additional weather verification

In order to assess whether the conclusions drawn above can be extended to other variables, we compute the RMSE values for four additional weather variables. Figures 6a, 6b, 6c and 6d show the RMSE values in function of lead time for geopotential height at 500 hPa (z500), specific humidity at 700 hPa (q700), temperature at 850 hPa (t850) and surface temperature (t2m). It becomes clear that there is no clear recurrent trend for all variables besides the underperformance of the GNN for early lead times. In the case of z500, for example, the Transformer performs significantly better at large lead times, whereas for t2m the opposite is true. This suggests that different models might be preferable for different variables and that no single model comes out as the best performing overall.



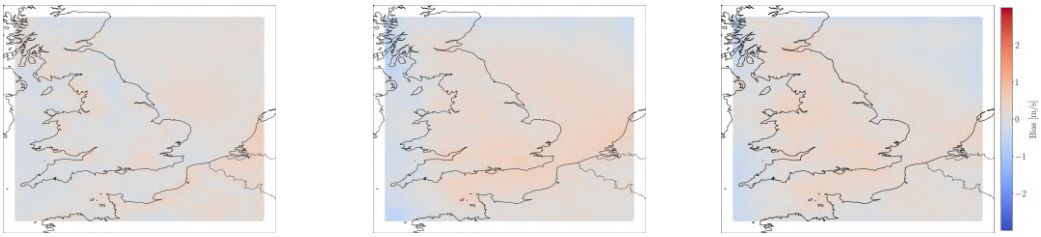
(a) z500 (b) q700

(c) t850 (d) t2m

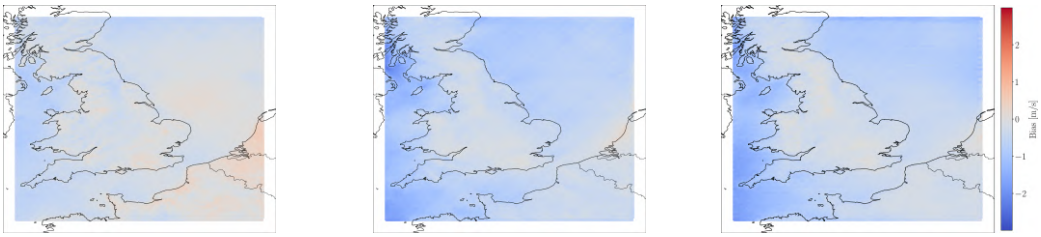
Figure 6: RMSE in function of lead time for all three models for geopotential height at 500 hPa (*a*), specific humidity at 700 hPa (*b*), temperature at 850 hPa (*c*) and surface temperature (*d*).

Figures 7 and 8 show the bias of all three models for ws10 and ws100 for multiple lead times. For all models, underestimation of wind speed at both heights becomes more significant with increasing lead times over the North Sea region, albeit more articulate for the GraphTransformer and Transformer than for the GNN. Underestimation of wind speed on the mainland on the other hand, reflected by red regions, becomes less significant for longer lead times.

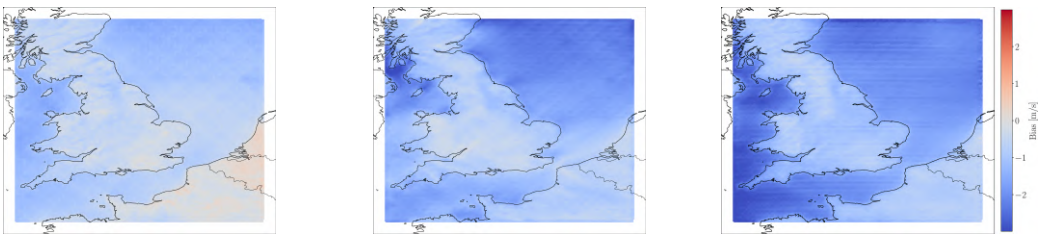
(a) GNN (3h) (b) GraphTransformer (3h) (c) Transformer (3h)



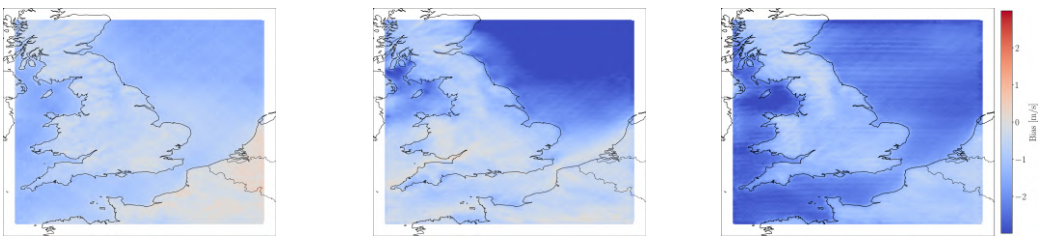
(d) GNN (12h) (e) GraphTransformer (12h) (f) Transformer (12h)



(g) GNN (24h) (h) GraphTransformer (24h) (i) Transformer (24h)



(j) GNN (48h) (k) GraphTransformer (48h) (l) Transformer (48h)



(m) GNN (72h) (n) GraphTransformer (72h) (o) Transformer (72h)

Figure 7: Forecast bias (forecast − observation) for wind speed at 10 meter for different lead times and model architectures. Columns correspond to models (GNN, GraphTransformer, Transformer); rows correspond to lead times.

B. Meteorological variables

The variables included in both the CERRA and ERA5 dataset are presented in Table 1 below.

Table 1: Input variables used in this study. Atmospheric variables are provided on multiple pressure levels (500, 600, 700, 750, 800, 850, 900, 950, 1000 hPa), while wind-related variables are available at 10 and 100 m. Wind direction and latitude/longitude are provided as sine and cosine components.

Surface	**Height**	**Atmospheric**
2 m temperature (t2m)	Wind speed (ws10, ws100)	Temperature (t)
Mean sea-level pressure (msl)	Wind direction (wdir10, wdir100)	U component of wind (u)
Surface geopotential (z_sfc)		V component of wind (v)
Medium cloud cover (mcc)		Geopotential height (z)
		Specific humidity (q)
Forcings		
Land-sea mask (lsm)	Insolation	Julian day (cos/sin)
Latitude (cos/sin)	Longitude (cos/sin)	

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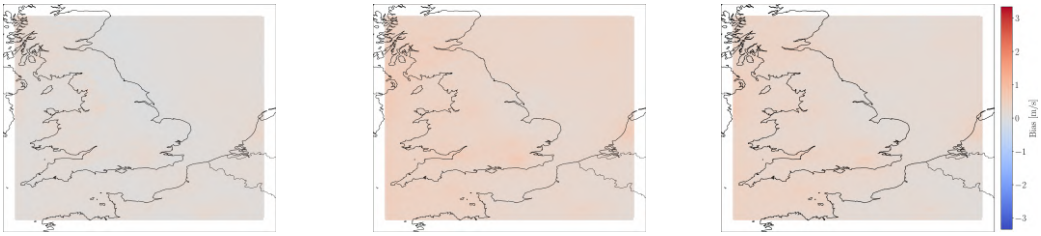
Competing Interests. None.

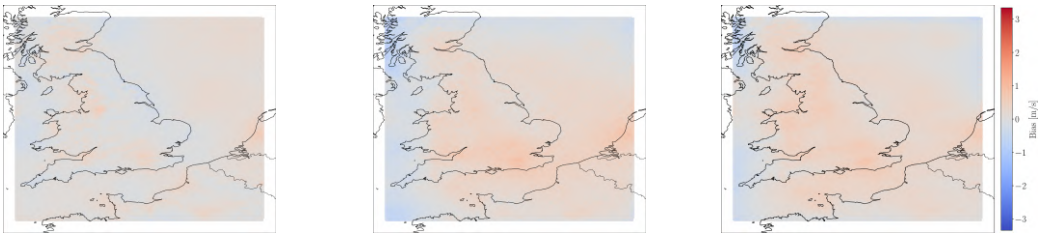
Data Availability Statement. All data and code can be freely downloaded via the public Github pages https://github.com/UAntwerpM4S/WindAI and https://github.com/UAntwerpM4S/WindPowerTransformer.

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

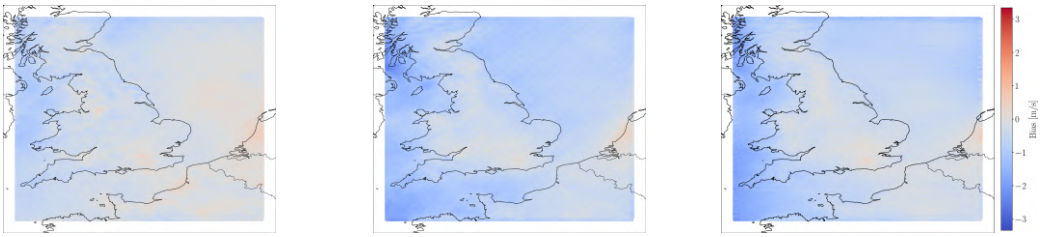
Author Contributions. **Aaron Van Poecke**: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization **Michiel Van Ginderachter**: Conceptualization, Methodology, Writing – review & editing, Visualization **Joris Van den Bergh**: Conceptualization, Methodology, Writing – review & editing **Geert Smet**: Conceptualization, Methodology, Writing – review & editing **Dieter Van den Bleeken**: Conceptualization, Methodology, Writing – review & editing, Visualization **Hossein Tabari**: Conceptualization, Methodology, Resources, Supervision **Peter Hellinckx**: Conceptualization, Methodology, Resources, Supervision



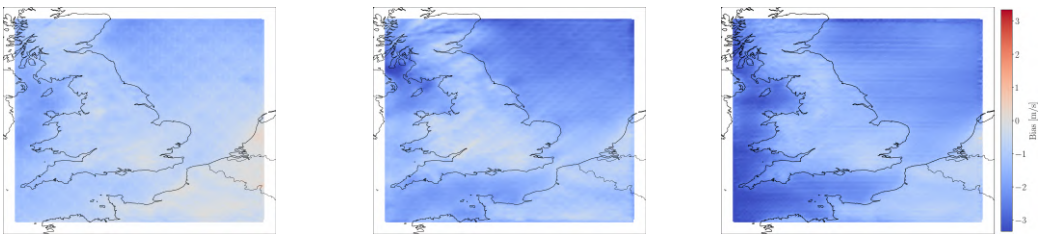
(a) GNN (3h) (b) GraphTransformer (3h) (c) Transformer (3h)



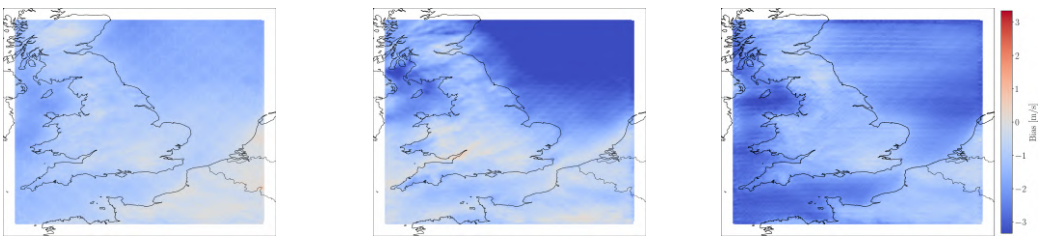
(d) GNN (12h) (e) GraphTransformer (12h) (f) Transformer (12h)



(g) GNN (24h) (h) GraphTransformer (24h) (i) Transformer (24h)



(j) GNN (48h) (k) GraphTransformer (48h) (l) Transformer (48h)



(m) GNN (72h) (n) GraphTransformer (72h) (o) Transformer (72h)

Figure 8: Forecast bias (forecast − observation) for wind speed at 100 meter for different lead times and model architectures. Columns correspond to models (GNN, GraphTransformer, Transformer); rows correspond to lead times.

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